

Cell Biology

MCQ

1. Grouping living organisms based on common features is called

- a) evolution
- b) classification
- c) reproduction
- d) adaptation

Answer: b) classification

2. About how many species of living organisms are identified till now?

- a) 1 million
- b) 3 million
- c) 5 million
- d) 8.7 million

Answer: d) 8.7 million

3. Biologists broadly classify living organisms into

- a) bacteria and fungi
- b) plants and animals
- c) unicellular and multicellular
- d) vertebrates and invertebrates

Answer: b) plants and animals

4. The method of arranging organisms into groups based on characteristics is known as

- a) evolution
- b) habitat
- c) classification
- d) variation

Answer: c) classification

5. Recognition of special features of objects is called

- a) assortment
- b) grouping
- c) identification
- d) classification

Answer: c) identification

6. Separating objects based on size, colour or shape is known as
- a) grouping
 - b) assortment
 - c) identification
 - d) taxonomy

Answer: b) assortment

7. Gathering similar items together after assorting is called
- a) grouping
 - b) identification
 - c) classification
 - d) sorting

Answer: a) grouping

8. A tool used to classify organisms based on differences is called
- a) microscope
 - b) taxonomy
 - c) dichotomous key
 - d) hierarchy

Answer: c) dichotomous key

9. In a dichotomous key, classification is based on
- a) many characters at once
 - b) one character at a time
 - c) habitat only
 - d) size only

Answer: b) one character at a time

10. A dichotomous key separates organisms based on characters that are
- a) similar
 - b) colourful
 - c) present or absent
 - d) inherited

Answer: c) present or absent

11. Animals are first classified based on the presence or absence of
- a) wings

- b) scales
- c) backbone
- d) legs

Answer: c) backbone

12. Animals with backbone are called

- a) invertebrates
- b) arthropods
- c) vertebrates
- d) molluscs

Answer: c) vertebrates

13. Animals without backbone are called

- a) mammals
- b) vertebrates
- c) invertebrates
- d) chordates

Answer: c) invertebrates

14. Aristotle classified organisms based on their

- a) food habits
- b) habitat
- c) size
- d) colour

Answer: b) habitat

15. Aristotle lived about

- a) 1000 years ago
- b) 1500 years ago
- c) 2000 years ago
- d) 2400 years ago

Answer: d) 2400 years ago

16. Aristotle classified animals into walkers, flyers and swimmers based on

- a) habitat
- b) reproduction
- c) movement

d) feeding

Answer: c) movement

17. Classification helps us to understand

a) only plants

b) only animals

c) similarities and differences

d) only habitats

Answer: c) similarities and differences

18. Things with similar characters are placed in

a) different groups

b) same group

c) random groups

d) mixed groups

Answer: b) same group

19. Classification helps in identifying organisms

a) incorrectly

b) partially

c) correctly

d) rarely

Answer: c) correctly

20. Classification helps to understand the

a) colour of organisms

b) evolution of organisms

c) height of organisms

d) weight of organisms

Answer: b) evolution of organisms

21. The arrangement of taxonomic categories in descending order is called

a) taxonomy

b) classification

c) hierarchy

d) nomenclature

Answer: c) hierarchy

22. The hierarchy of classification was introduced by

- a) Aristotle
- b) Whittaker
- c) Darwin
- d) Linnaeus

Answer: d) Linnaeus

23. The basic unit of classification is

- a) genus
- b) family
- c) species
- d) kingdom

Answer: c) species

24. How many main taxonomic categories are there?

- a) five
- b) six
- c) seven
- d) eight

Answer: c) seven

25. Which is the highest category in classification?

- a) species
- b) phylum
- c) class
- d) kingdom

Answer: d) kingdom

26. The five-kingdom classification was proposed by

- a) Linnaeus
- b) Aristotle
- c) R.H. Whittaker
- d) Darwin

Answer: c) R.H. Whittaker

27. The five-kingdom classification was proposed in

- a) 1950

b) 1960

c) 1969

d) 1975

Answer: c) 1969

28. Organisms without a well-defined nucleus belong to

a) Protista

b) Monera

c) Fungi

d) Plantae

Answer: b) Monera

29. Monera are

a) eukaryotes

b) multicellular

c) prokaryotes

d) autotrophs only

Answer: c) prokaryotes

30. Blue-green algae belong to the kingdom

a) Protista

b) Plantae

c) Monera

d) Fungi

Answer: c) Monera

31. Unicellular eukaryotic organisms belong to

a) Monera

b) Protista

c) Fungi

d) Animalia

Answer: b) Protista

32. Amoeba and Paramecium belong to

a) Monera

b) Fungi

c) Protista

d) Plantae

Answer: c) Protista

33. Fungi obtain food by

a) photosynthesis

b) ingestion

c) absorption

d) filtration

Answer: c) absorption

34. The cell wall of fungi is made of

a) cellulose

b) starch

c) chitin

d) protein

Answer: c) chitin

35. Mushrooms belong to the kingdom

a) Plantae

b) Fungi

c) Protista

d) Animalia

Answer: b) Fungi

36. Green plants are included in the kingdom

a) Fungi

b) Monera

c) Plantae

d) Protista

Answer: c) Plantae

37. Plants prepare their own food by

a) respiration

b) digestion

c) photosynthesis

d) fermentation

Answer: c) photosynthesis

38. Plants store food mainly in the form of

- a) protein
- b) glucose
- c) starch
- d) glycogen

Answer: c) starch

39. Flowering plants are also known as

- a) cryptogams
- b) gymnosperms
- c) phanerogams
- d) bryophytes

Answer: c) phanerogams

40. Plants with naked seeds are called

- a) angiosperms
- b) gymnosperms
- c) monocots
- d) dicots

Answer: b) gymnosperms

41. Seeds enclosed within fruits are found in

- a) gymnosperms
- b) algae
- c) angiosperms
- d) mosses

Answer: c) angiosperms

42. Monocot plants have

- a) two cotyledons
- b) no cotyledons
- c) one cotyledon
- d) many cotyledons

Answer: c) one cotyledon

43. Dicots have

- a) fibrous roots

- b) tap roots
- c) no roots
- d) adventitious roots

Answer: b) tap roots

44. Parallel venation is seen in

- a) dicots
- b) ferns
- c) monocots
- d) mosses

Answer: c) monocots

45. Non-flowering plants are called

- a) phanerogams
- b) angiosperms
- c) cryptogams
- d) gymnosperms

Answer: c) cryptogams

46. Algae have a body called

- a) stem
- b) root
- c) leaf
- d) thallus

Answer: d) thallus

47. Mosses are called amphibians of plant kingdom because they

- a) live in water only
- b) need moisture for life cycle
- c) have roots
- d) produce seeds

Answer: b) need moisture for life cycle

48. Ferns are known as

- a) non-vascular plants
- b) flowering plants
- c) vascular cryptogams

d) seed plants

Answer: c) vascular cryptogams

49. Animals belong to the kingdom

a) Protista

b) Plantae

c) Animalia

d) Fungi

Answer: c) Animalia

50. Animals lack

a) nucleus

b) cell wall

c) mitochondria

d) ribosomes

Answer: b) cell wall

51. Animals depend on others for food, hence they are

a) autotrophs

b) saprotrophs

c) heterotrophs

d) producers

Answer: c) heterotrophs

52. Animals without a body cavity are called

a) pseudocoelomates

b) coelomates

c) acoelomates

d) vertebrates

Answer: c) acoelomates

53. Animals with a true body cavity are called

a) acoelomates

b) pseudocoelomates

c) eucoelomates

d) invertebrates

Answer: c) eucoelomates

54. Symmetry where body parts are arranged around a central axis is

- a) bilateral
- b) asymmetry
- c) radial
- d) spherical

Answer: c) radial

55. Human beings show

- a) radial symmetry
- b) asymmetry
- c) bilateral symmetry
- d) no symmetry

Answer: c) bilateral symmetry

56. Sponges are examples of

- a) radial symmetry
- b) bilateral symmetry
- c) asymmetry
- d) spherical symmetry

Answer: c) asymmetry

57. Fish belong to the group

- a) invertebrates
- b) vertebrates
- c) protozoans
- d) molluscs

Answer: b) vertebrates

58. Starfish are

- a) vertebrates
- b) mammals
- c) invertebrates
- d) reptiles

Answer: c) invertebrates

59. One example of a phylum is

- a) Mammalia

- b) Chordata
- c) Aves
- d) Reptilia

Answer: b) Chordata

60. Binomial nomenclature gives each organism

- a) one name
- b) three names
- c) two names
- d) many names

Answer: c) two names

61. Binomial nomenclature was implemented by

- a) Aristotle
- b) Whittaker
- c) Linnaeus
- d) Darwin

Answer: c) Linnaeus

62. Father of Modern Taxonomy is

- a) Aristotle
- b) Darwin
- c) Linnaeus
- d) Whittaker

Answer: c) Linnaeus

63. The first name in binomial nomenclature is

- a) species
- b) family
- c) genus
- d) order

Answer: c) genus

64. The second name in binomial nomenclature is

- a) genus
- b) species
- c) family

d) class

Answer: b) species

65. Genus name should begin with a

a) small letter

b) capital letter

c) number

d) symbol

Answer: b) capital letter

66. Species name should be written in

a) capital letters

b) block letters

c) small letters

d) bold letters

Answer: c) small letters

67. Scientific names should be written in

a) red colour

b) bold

c) italics or underlined

d) brackets

Answer: c) italics or underlined

68. Taxonomy is the science of

a) breeding

b) classification and naming

c) evolution

d) nutrition

Answer: b) classification and naming

69. Nomenclature deals with

a) naming organisms

b) grouping organisms

c) evolution

d) habitat

Answer: a) naming organisms

70. One major merit of five-kingdom classification is that it is

- a) simple
- b) unscientific
- c) scientific and natural
- d) confusing

Answer: c) scientific and natural

71. Viruses are not properly placed in

- a) two-kingdom system
- b) five-kingdom system
- c) plant kingdom
- d) animal kingdom

Answer: b) five-kingdom system

72. A drawback of five-kingdom classification is

- a) ignores evolution
- b) includes viruses
- c) excludes plants
- d) places multicellular organisms arising many times

Answer: d) places multicellular organisms arising many times

73. Algae are mostly

- a) terrestrial
- b) aerial
- c) aquatic
- d) desert plants

Answer: c) aquatic

74. Protozoans are

- a) plant-like protists
- b) animal-like protists
- c) fungi
- d) bacteria

Answer: b) animal-like protists

75. Euglena belongs to

- a) Monera

- b) Protista
- c) Fungi
- d) Plantae

Answer: b) Protista

76. Yeast is an example of

- a) algae
- b) bacteria
- c) fungi
- d) protozoa

Answer: c) fungi

77. Ferns grow best in

- a) dry deserts
- b) snowy regions
- c) shady moist places
- d) oceans

Answer: c) shady moist places

78. Animals respond to stimuli due to the presence of

- a) muscles
- b) digestive system
- c) nervous system
- d) circulatory system

Answer: c) nervous system

79. Which group shows autotrophic nutrition?

- a) Animalia
- b) Fungi
- c) Plantae
- d) Protozoa

Answer: c) Plantae

80. Scientific naming avoids confusion because

- a) names are local
- b) names are universal
- c) names are complex

d) names are long

Answer: b) names are universal